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Bihar Engineering University, Patna

B.Tech 2nd Semester Examination, 2025

Time: 03 Hours

Code: 100206 ***Subject: Workshop Practices*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven question only): **[14]**

(a) The sheet metal gauge is used to measure
- (i) Length
- (ii) Thickness of sheet
- (iii) Hardness
- (iv) Angle

(b) The metal commonly used in sheet metal work is
- (i) Cast iron
- (ii) Tin plate
- (iii) Bronze
- (iv) Lead

(c) Soldering uses a Filler metal with melting temperature
- (i) Equal to base metal
- (ii) Above 1000°C
- (iii) Below 450°C
- (iv) Without heating

(d) In arc welding, heat is produced by
- (i) Gas flame
- (ii) Electric arc
- (iii) Chemical reaction
- (iv) Friction

(e) The material used for patterns is NOT
- (i) Wood
- (ii) Aluminium
- (iii) Brass
- (iv) Glass

(f) The tool used for ramming sand in moulding is
- (i) Trowel
- (ii) Slick
- (iii) Rammer
- (iv) Shovel

(g) Draw filing is used to
- (i) Cut threads
- (ii) Remove bulk material
- (iii) Mark lines
- (iv) Obtain smooth surface finish

(h) A common carpentry joint is
- (i) Lap joint
- (ii) Brazed joint
- (iii) Riveted joint
- (iv) Welded joint

(i) In smithy, heating of metal is done in
- (i) Furnace
- (ii) Lathe
- (iii) Shaper
- (iv) Drill press

(j) The forging operation used to increase length and reduce cross-section is
- (i) Upsetting
- (ii) Drawing (Drawing down)
- (iii) Punching
- (iv) Bending

Q.2 Solve both questions : **[14]**

(a) List any five sheet metal tools and describe their uses.

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(b) Explain the process of scribing and bending in sheet metal work.

Q.3 Solve both questions : **[14]**

(a) Explain the difference between brazing and soldering.

(b) Describe mechanical joints with examples.

Q.4 Solve both questions : **[14]**

(a) Describe the different pattern materials used in foundries.

(b) Explain the function of riser, runner, and sprue in sand casting with a neat sketch.

Q.5 Solve both questions : **[14]**

(a) Name and explain the types of files used in fitting shop.

(b) Describe sawing and chipping operations.

Q.6 Solve both questions : **[14]**

(a) Explain any three defects in wood.

(b) Describe carpentry measuring and marking tools.

Q.7 Solve both questions : **[14]**

(a) Define seasoning of wood. What are the different methods?

(b) Differentiate between hot working and cold working processes.

Q.8 Solve both questions : **[14]**

(a) Write down the procedure of the product development: hexagonal box with cap in sheet metal shop.

(b) What safety precautions should be considered and followed in a workshop during various operations?

Q.9 Write short notes on of the following: **[14]**

(a) Drawing
(b) Upsetting
(c) Bending
(d) Green Sand

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